Database ERD: Black Friday Platform

The PostgreSQL 16 database `fridayblack` (with pgvector), as read from the live database after Alembic migration `0004`.

1. Overview

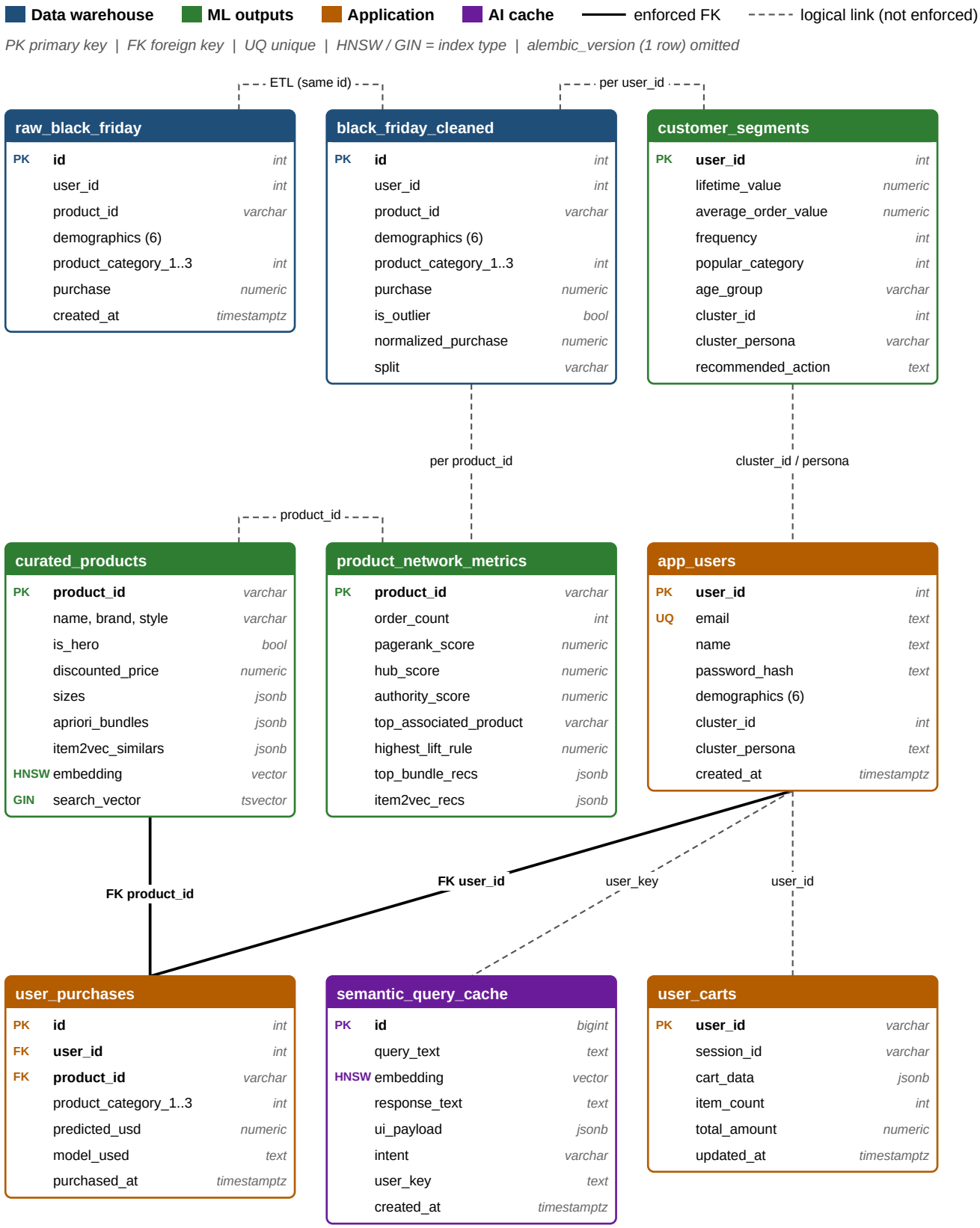
The database has **10 tables in 4 groups**:

Group	Tables	Written by	Rows (live)
Data warehouse	`raw_black_friday`, `black_friday_cleaned`	`make ingest`, `make preprocess`	550,069 each
ML outputs	`customer_segments`, `product_network_metrics`, `curated_products`	`make segmentation`, `make basket`, `seed_curated`	5,891 / 133 / 50
Application	`app_users`, `user_purchases`, `user_carts`	The FastAPI backend	1,644 / 127 / 0
AI cache and system	`semantic_query_cache`, `alembic_version`	The AI assistant, Alembic	0 / 1

Enforced vs logical relationships. Two relationships are enforced by real foreign keys, both on the application tables: `user_purchases.user_id` references `app_users.user_id`, and `user_purchases.product_id` references `curated_products.product_id` (migration `0003`). The others are **logical**: the columns match, but the database does not check them. That is normal for an analytics warehouse loaded in bulk, where foreign keys would slow down the 550k-row loads and block truncate-and-reload.

2. Entity relationship diagram

Solid line = enforced foreign key, dashed line = logical relationship.


Data warehouse
ML outputs
Application
AI cache
enforced FK
logical link (not enforced)
PK primary key | FK foreign key | UQ unique | HNSW / GIN = index type | alembic_version (1 row) omitted
ETL (same id)
per user_id
raw_black_friday
PK id int
user_id int
product_id varchar
demographics (6)
product_category_1..3 int
purchase numeric
created_at timestamptz
black_friday_cleaned
PK id int
user_id int
product_id varchar
demographics (6)
product_category_1..3 int
purchase numeric
is_outlier bool
normalized_purchase numeric
split varchar
customer_segments
PK user_id int
lifetime_value numeric
average_order_value numeric
frequency int
popular_category int
age_group varchar
cluster_id int
cluster_persona varchar
recommended_action text
per product_id
cluster_id / persona
product_id
curated_products
PK product_id varchar
name, brand, style varchar
is_hero bool
discounted_price numeric
sizes jsonb
apriori_bundles jsonb
item2vec_similars jsonb
HNSW embedding vector
GIN search_vector tsvector
product_network_metrics
PK product_id varchar
order_count int
pagerank_score numeric
hub_score numeric
authority_score numeric
top_associated_product varchar
highest_lift_rule numeric
top_bundle_recs jsonb
item2vec_recs jsonb
app_users
PK user_id int
UQ email text
name text
password_hash text
demographics (6)
cluster_id int
cluster_persona text
created_at timestamptz
FK product_id
FK user_id
user_key
user_id
user_purchases
PK id int
FK user_id int
FK product_id varchar
product_category_1..3 int
predicted_usd numeric
model_used text
purchased_at timestamptz
semantic_query_cache
PK id bigint
query_text text
HNSW embedding vector
response_text text
ui_payload jsonb
intent varchar
user_key text
created_at timestamptz
user_carts
PK user_id varchar
session_id varchar
cart_data jsonb
item_count int
total_amount numeric
updated_at timestamptz

On GitHub (and in VS Code with a Mermaid extension) this diagram is drawn from the Mermaid code. In the PDF version it is the full-page picture on the previous page.

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3. How the data flows

```
train.csv
  |
  |  make ingest (COPY, ~7s)
  v
raw_black_friday
  |
  |  make preprocess (MissForest imputation, outliers, train/test split)
  v
black_friday_cleaned
  |
  +--> make segmentation (Gower clustering) --> customer_segments (10 personas)
  |
  +--> make basket (Apriori, Item2Vec, PageRank/HITS) --> product_network_metrics
                                                                |
                                                                |  seed_curated
                                                                v
                                                          curated_products


FastAPI backend (reads the tables above, writes the application tables):

  app_users  --(signup: demographics + persona from customer_segments)
     |
     +-- user_purchases   (FK user_id; ONNX price prediction per purchase)

  user_carts             (durable cart; Redis is the fast tier)
  semantic_query_cache   (AI assistant answers, per user_key)
```

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4. Tables in detail

4.1 raw_black_friday (data warehouse)

The original transactions, exactly as in `train.csv`. **Never modified after loading.**

Column	Type	Null	Notes
`id`	integer	no	**PK**
`user_id`	integer	no	Dataset customer id (indexed)
`product_id`	varchar(32)	no	e.g. `P00025442` (indexed)
`gender`, `age`, `occupation`, `city_category`, `stay_in_current_city_years`, `marital_status`	varchar / int	no	Customer demographics
`product_category_1`	integer	no	Indexed
`product_category_2`, `product_category_3`	integer	**yes**	Missing in the raw data
`purchase`	numeric(12,2)	no	Amount in INR (the regression target)
`created_at`	timestamptz	yes	Load time

4.2 black_friday_cleaned (data warehouse)

Same rows as raw, after cleaning: categories 2 and 3 **imputed** by MissForest, outliers flagged, target normalised, and a train/test split assigned.

Column	Type	Null	Notes
`id`	integer	no	**PK**, same id as the raw row
`user_id`, `product_id`	integer / varchar(32)	no	Indexed
demographics + `product_category_1..3`		no	Categories 2 and 3 are now **never null** (imputed)
`purchase`	numeric(12,2)	no	
`is_outlier`	boolean	no	Indexed; outlier rows are excluded from drift batches
`normalized_purchase`	numeric(8,6)	no	The model's training target
`split`	varchar(16)	no	`train` or `test` (indexed); fixed before any fitting, so no leakage

4.3 customer_segments (ML output)

One row per dataset customer, with RFM-style features and their cluster.

Column	Type	Notes
`user_id`	integer	**PK** (dataset customer id)
`lifetime_value`, `average_order_value`	numeric	Spending features (LTV indexed)
`frequency`	integer	Distinct products bought (PIPE-13 explains why it isn't a purchase count)
`purchase_amount_variability`, `popular_category`	numeric / int	
`gender`, `marital_status`, `age_group`, `age_binned`	varchar	
`cluster_id`	integer	0–9 (indexed)
`cluster_persona`, `recommended_action`	varchar / text	One of the **10 personas** and its marketing action

4.4 product_network_metrics (ML output)

Graph and recommendation scores per product, from Apriori rules and the product co-purchase network.

Column	Type	Notes
`product_id`	varchar(32)	**PK**
`order_count`	integer	
`pagerank_score`, `hub_score`, `authority_score`	numeric(8,5)	Network centrality (each indexed)

Column	Type	Notes
`top_associated_product`, `highest_lift_rule`	varchar / numeric	The best Apriori rule
`top_bundle_recommendations`, `item2vec_recommendations`	jsonb	Lists of recommended products

4.5 curated_products (store catalog)

The 50 products shown in the store, enriched with names, images, prices and recommendations.

Column	Type	Notes
`product_id`	varchar(32)	**PK**
`name`, `title`, `tagline`, `description`, `brand`, `style`, `season`, `badge`	varchar / text	Display text
`category_name`, `category`, `gender`	varchar	Used by the sidebar filters
`is_hero`	boolean	The featured product on the right of the page
`original_price`, `discounted_price`	numeric(10,2)	Prices shown in the UI, in USD (hand-maintained in `data/curated_products.json`)
`order_count`	int	Purchase rows in the warehouse (rebuilt by `make catalog`)
`sizes`, `apriori_bundles`, `item2vec_similars`	jsonb	Sizes and "bought together" / "similar" lists (bundles and similars rebuilt by `make catalog` from the Apriori rules and the Item2Vec model)
`embedding`	vector	**pgvector** embedding for semantic search (**HNSW index**)
`search_vector`	tsvector	Full-text search (GIN index)

4.6 app_users (application)

Registered store users. Separate from the dataset customers in the warehouse: `app_users.user_id` is a store account id, `customer_segments.user_id` / `black_friday_*.user_id` are dataset customer ids (see section 6, item 5).

Column	Type	Notes
`user_id`	integer	**PK**
`email`	text	**UNIQUE** (indexed); stored lower-case
`name`, `password_hash`	text	bcrypt hash, never the password itself
`gender`, `age`, `city_category`, `occupation`, `marital_status`, `stay_in_current_city_years`		Demographics, used as model inputs for personal price predictions

Column	Type	Notes
`cluster_id`, `cluster_persona`, `recommended_action`		The persona assigned at signup
`created_at`	timestamptz	

4.7 user_purchases (application)

Column	Type	Notes
`id`	integer	**PK**
`user_id`	integer	**FK to** `app_users.user_id`, not null (indexed)
`product_id`	varchar(32)	**FK to** `curated_products.product_id` (indexed): the product bought
`product_category_1..3`	integer	
`predicted_usd`	numeric(10,2)	The member price at purchase time, from the ONNX prediction
`model_used`	text	Which model made the prediction
`purchased_at`	timestamptz	

4.8 user_carts (application)

Durable cart storage ("cold tier"; Redis is the fast tier).

Column	Type	Notes
`user_id`	varchar(64)	**PK**: JWT subject, or a `guest_<hex>` id
`session_id`	varchar(64)	Browser session
`cart_data`	jsonb	Cart items
`item_count`, `total_amount`	int / numeric	
`updated_at`	timestamptz	Last save (default `now()`)

4.9 semantic_query_cache (AI)

Caches the assistant's answers by meaning, so similar questions reuse an answer.

Column	Type	Notes
`id`	bigint	**PK**
`query_text`, `response_text`	text	The question and the answer
`embedding`	vector	The question's embedding (**HNSW index** for nearest-neighbour search)
`ui_payload`	jsonb	Product cards shown with the answer
`intent`	varchar(64)	The router's intent label
`user_key`	text	Owner of a personalised answer, so it isn't served to someone else (CORE-15)

Column	Type	Notes
`created_at`	timestamptz	Indexed together with `user_key`

4.10 alembic_version (system)

One row with the current migration version (`0004`). Managed by Alembic only; never edit it by hand.

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5. Special column types

Type	What it is	Used for
`jsonb`	Binary JSON: flexible nested data that can still be queried and indexed	Bundles, recommendations, cart contents, UI payloads
`vector`	**pgvector**: a list of numbers (an embedding)	Semantic search and the semantic cache
HNSW index	A fast **approximate nearest-neighbour** index for vectors	"Find the most similar product or question"
`tsvector` + GIN index	PostgreSQL full-text search	Keyword search in the catalog
`numeric(p,s)`	Exact decimal (p digits, s after the point)	Money and scores; no floating-point rounding errors
`timestamptz`	Timestamp with time zone	Audit times

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6. Observations

These were noticed while drawing this ERD. Round 2 (step 6.5) handled each one. Migration `0004` also dropped `curated_products.rating` / `review_count`: the dataset has no ratings, the values were invented, and nothing showed them.

1. **Money as `double precision`** in `curated_products` and `user_purchases.predicted_usd`. Fixed: `numeric(10,2)` (migration `0003`; every stored value already had at most 2 decimals). The invented `99.9` / `49.9` column defaults are gone, so a product without a price is an error instead of a made-up price.
2. **`user_carts.updated_at` was a `varchar`.** Fixed: `timestamptz` with default `now()`. Old ISO-8601 text converts; anything unreadable becomes NULL (never guessed). The repository now writes `now()`.
3. **Size mismatches with the warehouse.** Fixed: `app_users.gender` `varchar(8)`, `city_category` `varchar(4)`, `user_purchases.product_id` `varchar(32)`, the same as the warehouse.
4. **Only one foreign key.** Added `user_purchases.product_id` → `curated_products.product_id` (the store only sells catalogue products; the migration refuses to run while orphan rows exist). No foreign keys on the warehouse tables: they are bulk-loaded with TRUNCATE + COPY.
5. **Two kinds of "user".** `customer_segments.user_id` and `black_friday_*.user_id` are **dataset customers** (from `train.csv`); `app_users.user_id`, `user_purchases.user_id` and `user_carts.user_id` (a JWT subject or a `guest_<hex>` id) are **store accounts**. A store account is linked to a persona through `app_users.cluster_id`, never through `user_id`. Not renamed: the column name is used across the pipelines, the API and the tests, and a rename would gain only clarity, which this note gives.